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Let's Talk About Science:

A road to Science and Technology Studies

Frontiers of Physics

- Part II: Beyond Philosophy
  04/03/22



Where are we going today?

Beyond PoS
- Recap yesterday
- Underdetermination
- Kuhn and Paradigms

Paradigms
- Incommensurable contests
- Metaphysics in Science
- Practice of Science

STS
- Investigations of Science
- Criticisms & Errors
- Science Wars

Summing up
- Images and Implications
- Summing up
- Beyond (only) Empiricism

Recap: PoS

The story so far…

Looking for a *empirical method* in Science to justify high status

- *Ontological authority* from *epistemic privilege*

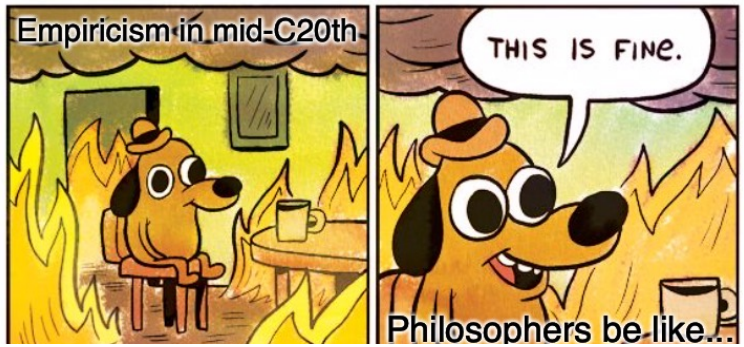
- Deduction does not do enough: need *amplicative* inference

- *Inductivism* seems relevant but incomplete
  - Need additional principles to read data and guide inferences
  - Some reason to think these are not *empirical*

- *Hypothetico-Deductive* method moves *empirical* aspect to testing
  - Invention of a theory is separate from justification
  - Confirmation has problems with how to deal with trivial agreement
- Falsification saw limitations of agreement
  - Use falsification as a guide for progress
  - Get *demarcation* principle back

Empiricism seemed a viable justification

But Wait!



Observation & Theory

Two philosophers raised issues with this picture at the start of the 20th century

- Breaking the empirical connection between observation and theory

Pierre **Duhem**:

- Experiments do not test a single theory in isolation
- Many auxiliary theories are included: apparatus, analysis, context

When experiment and theory disagree, there are *alternative* sources of blame

Willard Van Orman **Quine**:

- The result of any experiment has (logically) an infinite number of interpretations

As a point of logic, you can always add extra information so long as the observation is not contradicted

This is now known as the **Duhem-Quine thesis**

Theoretical knowledge is *underdetermined* by experimental observation

- Additional information is required: background theories, preferences
- Difficult to *demarcate a priori* when this is acceptable science or not

Developing D-Q

Modern studies of science flesh out D-Q in practices to include modern distinctions

- Theoretical hypotheses: main/auxiliary hypotheses, principles of inference
- Phenomenology: techniques of transforming theory to prediction
- Apparatus: data-taking machinery
- Analysis: techniques transforming “raw” data to measurement

Reaction to experimental disconfirmation/falsification

- Change/blame any of the above elements

Pickering called it the “Mangle of Practice”:

- Must overcome *resistance* of all the pieces so that they *accommodate* each other

For example: if a student does not observe $V=IR$ in a laboratory session *Ohm* is not disproven



Wrinkles be gone

Some principles are required to deal with the vast range of possible interpretations – **these are not empirically discovered**

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Paradigms

Thomas Kuhn (1922-1996)

Kuhn moved analysis of Science away from describing how an *ideal* Science might work and on to the *actual* History of Science

For Kuhn**, Science consists of sets of hypotheses used to solve problems –** *Paradigms*, e.g. Copernican solar system, Quantum Mechanics

- Includes theories to make empirical predictions and the required assumptions stated or implicit
- Also, unspecified practices (perhaps now lost in history)



Anomaly don't care

Anomalies arise when data doesn't fit the predictions
but anomalies don't necessarily end (falsify) the paradigm

- Several strategies available to keep hypotheses
  - Dismiss test / change auxiliaries / change theory

Kuhn argued *history* shows:

- too many **anomalies inspire** new *Paradigms* - people will go elsewhere
- there is **no rule** to predict when people change *Paradigm*
- Scientists will find ***a way*** to decide winner between competitors

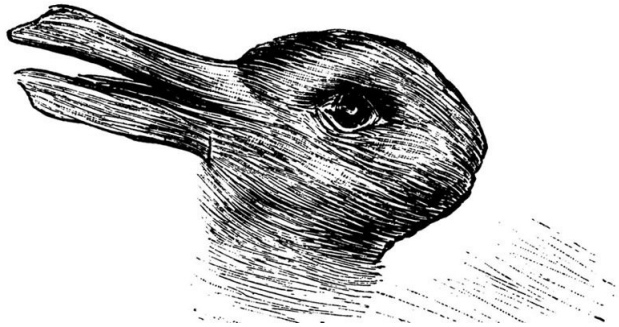
Changing Views

A *Paradigm* will have ways of obtaining, arranging, analysing data according to theory and practice

There is no guarantee observations are respected across *Paradigms*

- Revolution for Kuhn is a big thing – comparable to gestalt shift

How can *Paradigms* be compared? – some criteria…



Copernican **R**evolution
Quantum **M**echanics

	CR	QM
Predictive accuracy	Y	Y/N
Consistency	N	N
Scope	Y	N
Simplicity	Y(eventually)	Y
Fruitfulness	Y	Y

Kuhn: "I am suggesting, of course, that the criteria of choice with which I began function **not as rules**, which determine choice, **but as values**, which influence it."



If there are no fixed rules to compare paradigms (*incommensurable*) **what about *rational and cumulative* Science?**

Take a moment…

Incommensurability is a feature of many disputes



Consider some controversial debate

- Scientific: there's long history
- Non-scientific: politics, religion, sport, …

Can you identify some incommensurable points?

- Talk at cross purposes
- Contrasting interpretations of phenomena

Perhaps these disputes are not resolvable by rational argument alone

A Political Example

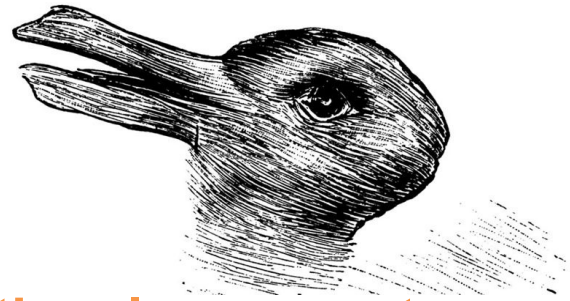
Paradigms are not peculiar to science. Example: Brexit

In times of *crisis* fixed concepts become *fractured* and *controversial*
e.g. Leave Vs Remain ca. 2019

Democratic Principle: which *votes* should be respected
- Leave: respect the referendum
- Remain: illegitimate referendum as ill-informed and unclear

Sovereignty of People: where is *The Will of the People*?
- Leave: Direct vote of referendum ca. 2016
- Remain: Representative Parliament from general election ca. 2017

Democratic Government: who are the *decision makers*?
- Leave: Executive Government (cabinet of major party)
- Remain: Representative Parliament (all MPs)



Conventions become contentious and common understanding dissolves

The "resolution" of the controversy may not be a rational argument

A Physics Example

Paradigms in examples can be found in Science: **Quantum Revolution**

In times of *crisis* fixed concepts become *fractured* and *controversial*
e.g. QM Vs Classical Physics ca. 1920s

Ontology: what is the fundamental *material* of physics
- Classical: point-like particles (simultaneous position and momentum)
- QM: wave-particles (Heisenberg's Uncertainty Principle)

Explanatory power: how to describe interactions
- Classical: complete particle trajectory over space-time
- QM: depends on experiment, not always possible (electrons and double slits)

Accuracy: precision of measurement/prediction
- Classical: in principle infinite for individuals, though limited by apparatus
- QM: in principle limited, only possible for distributions

Conventions become contentious and common understanding dissolves

So, what about *rational*, *singular*, *cumulative* Science?

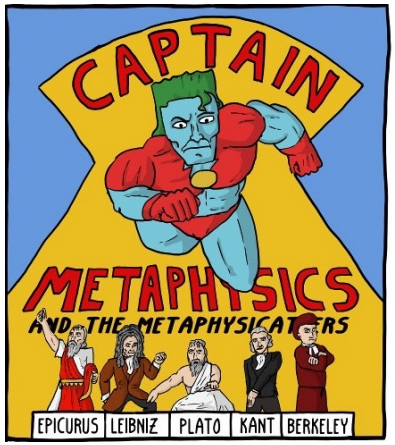


"Beyond Physics"

In many debates there are implicit assumptions (quietly) at work

Sometimes "natural" assumptions are not even mentioned:

- Every event has a cause
- Material objects persist without observation
- Nature is uniform – see part I
- Natural properties are not time dependent – see part I

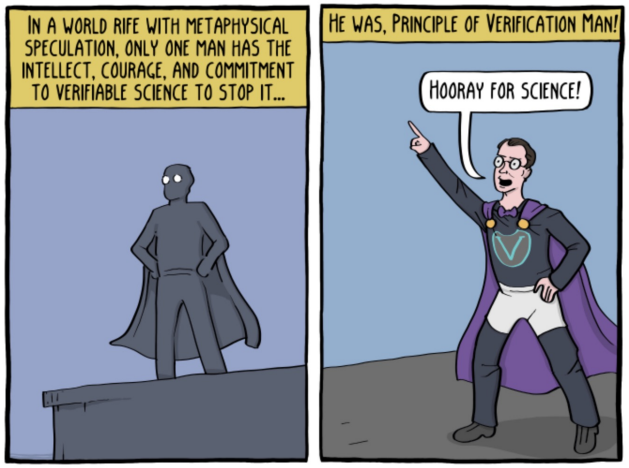


https://existentialcomics.com

Metaphysics is part of philosophy which investigates **assumptions that are (often implicit) in cognitive models** (a.k.a. theories of the world)

Not too popular in Philosophy of Science in early C20th

- *Verificationists* saw no place for metaphysics in science: what sort of knowledge is untestable?
- Most philosophers of science were at least suspicious



For Kuhn, **metaphysics is an untestable and indispensable part of any *Paradigm***

Science Studies

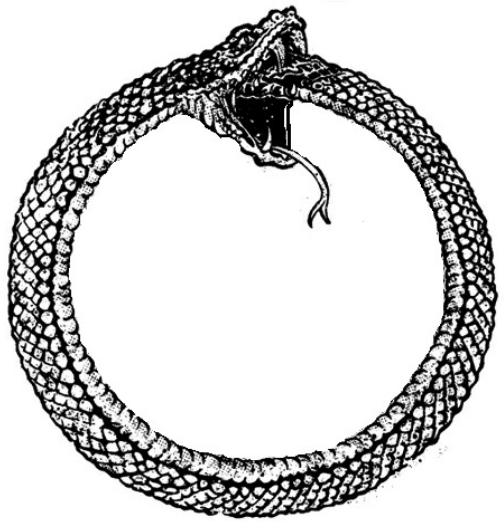
The idea that Science consists of *Paradigms* and *changes* in important ways between *Paradigms* has consequences for investigations of Science

Investigate science as *practiced* now and in the past

- Previous Paradigms: periods of “normal” science
- Controversies: periods of dispute and revolution (Popper’s interest)

Sources in the present: use methods of **Anthropology**

- *Participants*: researchers, reviewers, (non-)experts
- *Public Discussions*: meetings, talks
- *Practices*: tools and techniques



Sources in the past: use methods of **History**

- Remaining participants
- Recorded documents (mainly textual)
- Artefacts

In general, **this is an *empirical* study of people and their ideas**

- Application of science to itself: a systematic study of systematic study

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The Practice of Science

What is the **practice/culture/activity** of science?

- What do the people recognised as Scientists actually do?

More questions:

- What part of it is unique to things called Science?
  - Can this be used for *demarcation*
- Is it universal across all Sciences?
  - Are different practices appropriate for different contexts (*pluralism*)?
- Can it be replicated or removed?
  - How can new subjects join or be dismissed?

These questions raise issues involving

- *activity* of science – not (just) written histories of theories
- *authorities* in science – who makes the decisions

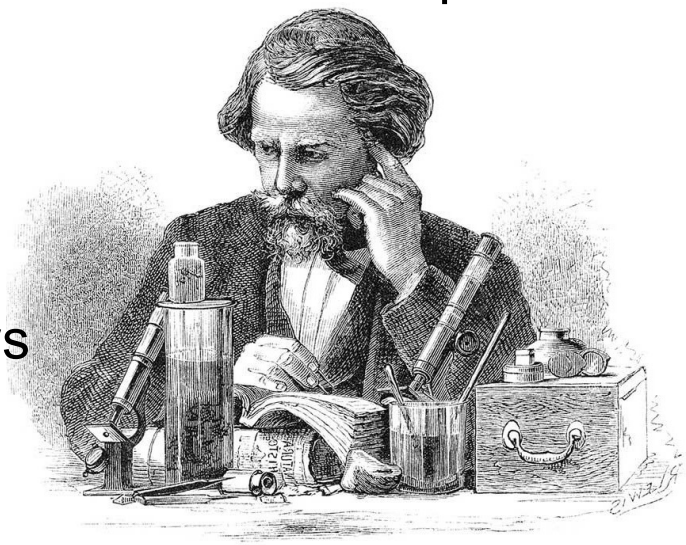
} **Empirical** task, Not **Analytic**

This is the task taken up by Science and Technology Studies

Take a moment…

Likely you have been trained to think of science as a history of ideas

- A series of theories punctuated by crucial, decisive experiments
- The scientist (often mid-aged white man) proposes and nature disposes
- But when was the last time you saw this guy?



This is an alternative way to consider science

- Series of Paradigms and crises
- Groups of scientists moving between world-views
- And these guys are on TV!

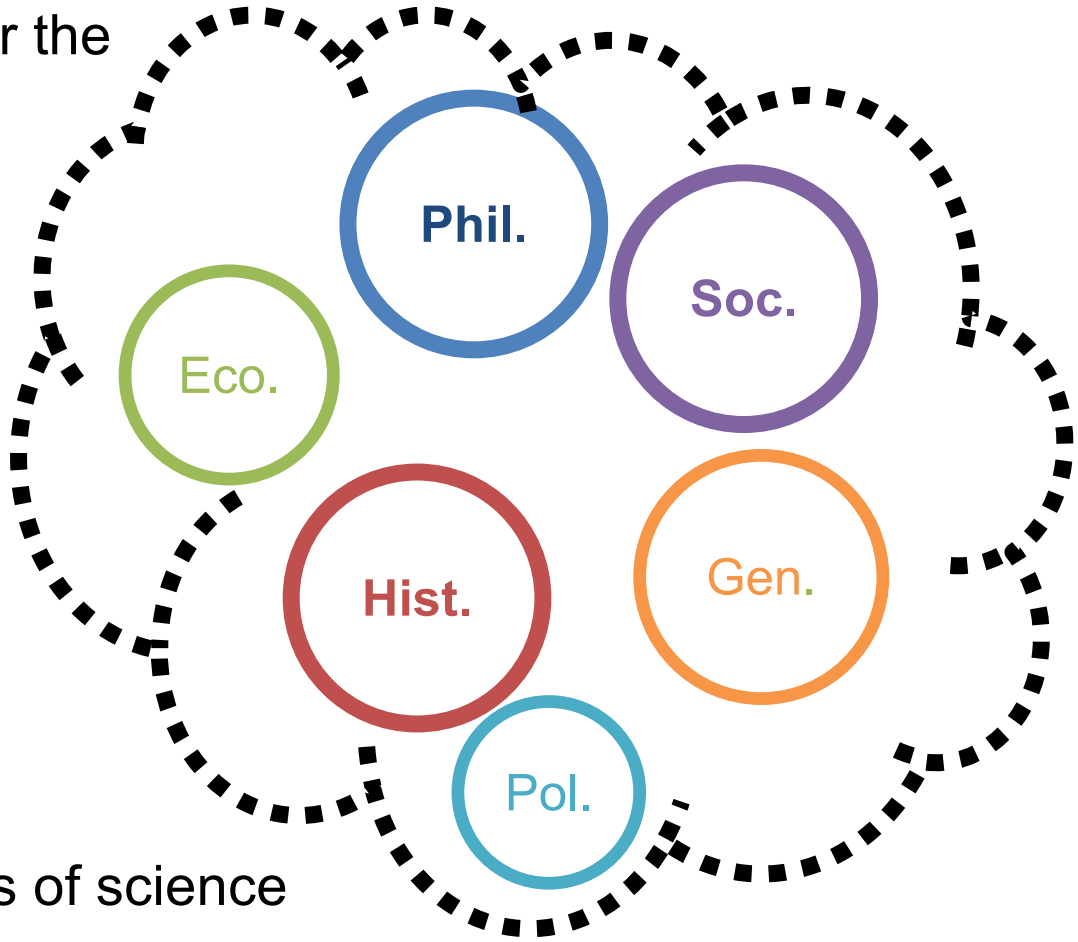


Reality Check

We have now moved to investigate the *practice* of science (not a *method*) to illuminate ontological authority

Science and Technology Studies

Lots of (sub-)subjects may fall under the banner of STS: 3 reviewed here



Philosophy (yesterday)
Traditionally a *normative* study:
Recommends how things should be

Sociology (today)
Traditionally a *descriptive* study:
Describes how things are

History (all over)
Resource of lessons and examples:
Inspiration and testbench of theories of science

NB Critique doesn't have to be critical: this isn't anti-science!

- Cf. film reviews, sports commentary, political pundits

"Old School" Sociology

During the 20th century scholars investigated human organisations including Religion, Health, Crime and … Science

Original Sociological Studies of Science conducted by Merton:

- ***Institutional environment***: university/laboratory/political climate
  - Effects on rate of progress of science, *but not on substance*
- ***Norms***: guiding principles to scientific activity
  - *communism*: sharing knowledge through published results
  - *universalism*: impersonal criteria of evaluation
  - *disinterestedness*: exclude personal interest
  - *organised scepticism*: (internal) criticism encouraged

By the 1970s this approach was out of fashion

- Limited studies to static Scientists (bodies in rooms)
- Ignored *content* of Science (theory and practice)



Outdated

Ludwig Fleck: **"[Previous studies] exhibit an excessive respect, bordering on pious reverence for scientific facts."**

The "New School"

The "New" Sociological Studies of Science (cf. 1960s+) wanted to go beyond the *macro*-sociological studies and investigate *micro*-sociological detail

- Moving from studies of Scientific institutions to the *practice of Science*

They did not accept the *Norms/Values* Scientists ascribed themselves

- The point of the empirical study is *find out* what happens *in practice*

Part of the *practice of Science* is deciding *Scientific facts*, i.e. the *content* of Science

- **Part of the "New" school was the Sociology of Scientific Knowledge (SSK) – how facts are decided**

Some people worried this would lead to sociologists doing bad science

- For example: it's not a sociologist's place to decide facts of Physics

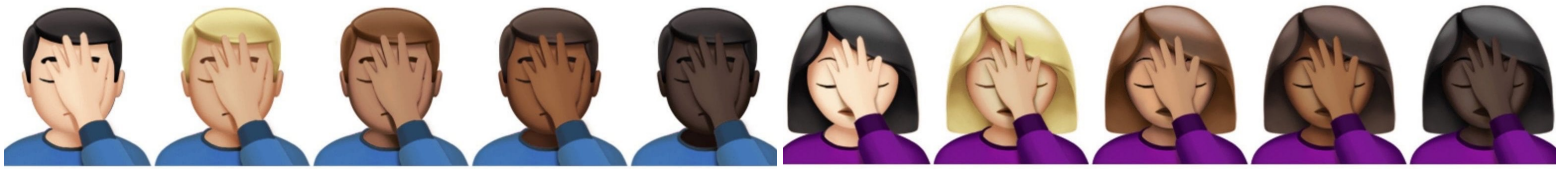
Sociologists argued they were not interested in the outcome of decisions but how the decisions were made

They also complained scientists had been doing bad sociology for years…

Error Analysis

Interest in practitioners of Science has been around for a long time

- but old stories about Scientific failure tend to focus on individuals



Sociology of Scientific Error: people only matter when something breaks

History used to supply anecdotes with pedagogical use:

- *Just-so stories*: (legendary) explanation for a practice/behaviour
  - E.g. Motions of the planets (ignore unfashionable reasoning)
- *Morals and Fables*: How to be a good scientist
  - E.g. Stand on the shoulders of giants (ignore personal faults)
- *Lessons learnt*: Illuminating mistakes with celebrated solutions
  - E.g. The unnecessary Ether (ignore historical detail)

These are not *empirical* investigations of the practice of Science

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Take a moment…

Sociology of Scientific Error: people only matter when something breaks

- Just-so stories: Untestable explanation for a practice/behaviour
  - E.g. Motions of the planets
- Morals and Fables: How to be a good scientist
  - E.g. Stand on the shoulders of giants
- Lessons learnt: Illuminating mistakes with celebrated solutions
  - E.g. The unnecessary Ether

During your time as a student of physics/science what role have people played in the stories: heroes, fools, geniuses

- What was the importance of that role?
  - Hampering science? Leaping forward?

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SSK Schools

Sociology of Scientific Knowledge (SSK) was interested in how Scientific facts came to be recognised

Various approaches:

Edinburgh/Bath school's Strong Program

- **Methodological Relativism**: Explanation of success/failure cannot rely on retrospective "truth" of a theory

Andrew Pickering

- **Contingency**: If people hadn't made certain decisions or accidents then the theories would have been different

Paris

- **Actor-Network Theory**: Reject theory-laden language to describe science: e.g. abandoned distinctions between natural/social, actor/object

These are not arguments about how to do Science
These are arguments about how to describe/understand Science

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Science Wars

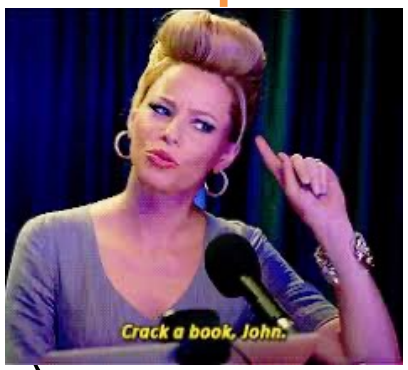
By this point (1990s) Scientists were starting to hear whispers on campus …

- Historians talking about science: Leviathan and the Air-pump
- Philosophers talking about science: Structure of Scientific Revolutions
- Sociologists talking about science: Laboratory Life
- Plus others: gender studies, social theory, economists, religious figures

Most scientists didn't know/care, but some decided to "defend" science from attack – self-styled 'Guardians of Science'

- Sides were mixed but "guardians" were mostly physicists
- Spat began in academia and expanded into newspaper articles

Summation: two sides of trained experts arguing beyond their expertise (Lynch)



Mostly polemics with poor standard of academic debate

- Arguments focussed on philosophical "sticking points"
- Rarely were philosophers involved
- Rarely were previous debates cited (2000+ years of debate)

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Science Wars II

Hacking's analysis of the "Sticking Points" at issue:

1. *Contingency* of Scientific Knowledge: Could Science look different?
    - If so, how different?
    - Could the *content* of scientific theories be different?

2. *Nominalism*: What is the status of Scientific facts, laws, etc.
    - Objective representation of the world, organisation of phenomena, useful rules for application?

3. *Stability*: In what sense is science stable as it develops
    - Changes driven by internal factors alone or external factors?
    - Is Science a continuous line of progress or a series of *Paradigms*?

The arguments were generally **not** well conducted and loudest voices often least measured

- **The fever of the debate peaked with *The Sokal Affair***
    - As close as you can get to fame/infamy in STS



Sokal Affair

Actors

- NYU physics professor *Alan Sokal*
- Academic journal (postmodern cultural studies) *Social Text*
- Academic magazine *Lingua Franca*

Events

- *Sokal* wrote a paper published in *Social Text*

"Transgressing the Boundaries: Towards a Transformative Hermeneutics of Quantum Gravity"

-- summary: Quantum gravity is a social and linguistic construct

- A few weeks later Sokal told *Lingua Franca* the paper was a hoax(!)

Issues

- Sokal claimed he wanted to see if he could

  "publish an article liberally salted with nonsense if *(a)* it sounded good and *(b)* it flattered the editors' ideological preconceptions"

- Critics said he broke the conventions of academia and showed nothing

Outcome

- Sympathists (mostly physicists) sniggered
- Antagonists shook their heads
- Outsiders were bewildered



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Take a moment…

What do you think about these issues

1. *Contingency* of Scientific Knowledge: Could Science look different?
    - If so, how different?
    - Could the *content* of scientific theories be different?

2. *Nominalism*: What is the status of Scientific facts, laws, etc.
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3. *Stability*: In what sense is science stable as it develops
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Your reaction may give you some idea of your attitude to Science

- This doesn't have to be your final answer

A meta-for Science

You may have an (implicit) image of Science that guides your thinking on STS issues: *Contingency, Nominalism, Stability*

- A principle that guides but is not necessarily testable

What's your *metaphor* of Science/Reality?

- Jigsaw Picture
  - Topics as pieces
  - Discrete pieces fit together
  - Goal: full, accurate picture of reality
- Mesh of theories
  - Topics as regions
  - Some parts more detailed than others
  - Goal: comprehensive, coherent, fine mesh
- Cloud of theories and practices
  - Diffuse, loosely ordered
  - Goal: is use as required



Imagery & Implications



A strong (Realist) image of Science not only helps justify a privileged position in society but also an authoritative role

Authority of definition
- What counts as (Scientific) Knowledge

Authority of policing
- Defining boundaries of (Scientific) Knowledge
- Defining quality of (Scientific) Knowledge



Authority over dissent
- What counts as legitimate criticism?
- Who gets to criticise?

→ Added to an idea of exceptionalism this becomes a monopoly
  - Only *real* knowledge is *Scientific* knowledge

STS aims to get *beyond the metaphor* and to what Scientists *do*

Re: Experience to Knowledge

Steps to *Scientific* knowledge:



There is a long history of attempts to ground authority of Science in this way

- Focus on empiricism to justify authority
  e.g. Inductivism, Hypothetico-Deductive method, …

We saw (arguably) in these methods empiricism was mixed with metaphysics

- Cf. criticisms of Hume, Goodman, Hempel, Popper, Kuhn

This is a problem when Science is conceived as *only* an *empirical* method

- Worse when a single (realist) view is the measure of all views

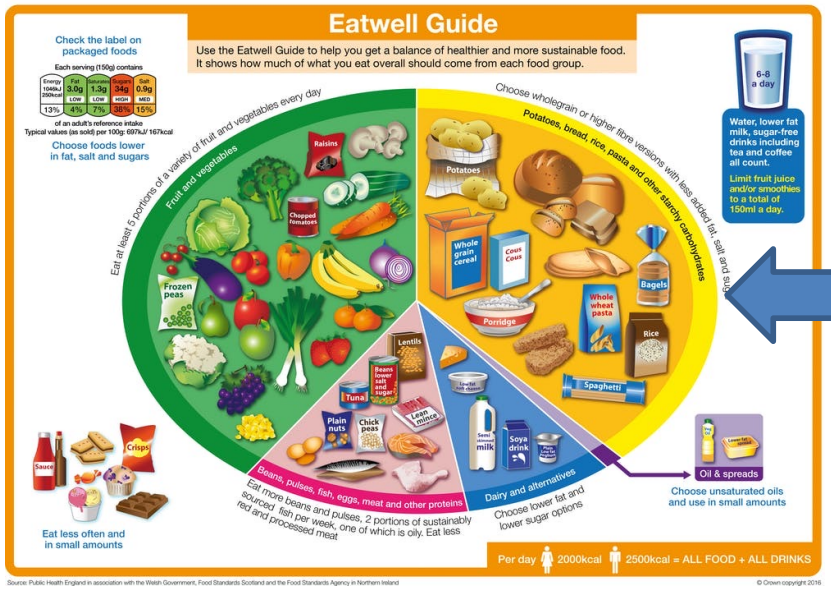
An alternative is to see metaphysics as intrinsic part of *Science*

- Principles of inference, practices with data, values of theory comparison

More Summing-up

Identifying *Science* as *Empirical Knowledge* is like marketing whisky as *wheat-based nutrition*

- There is truth to it but it's missing much of the interesting features





To extend the metaphor further…

Empiricism is part of a healthy knowledge diet: look to the world!

But an Empiricism only diet is malnutrition

- Make sure to get your meta-physics: the guiding principles to help inference from experience along

Beyond (only) Empiricism

What are the implications if the focus on empiricism is diminished?

- Depends on how fixed the view of science is
- Change of view: trees → woods



Unbiased: Based on observed data, not innate prejudice

- **Bias cannot be expelled but it can be shared**

Objective: Based on public data, not subjective opinion

- **Commonly held (inter-subjective) experience**

Cumulative: Based on growing data, improves over time

- **Cumulative but in context, and we live in context**

Autonomous: Based on internal data, not external influences

- **Nothing is autonomous, and that's ok**

This is not an overhaul of *Scientific Practice;* this is a change in PoV

- No longer about interrogating a *singular nature*
- Now about **shared** *agreement*, **common** *understanding* and **context** of (*Scientific*) *knowledge* … between ***people***

Caveat (again)

WARNING

Almost everything mentioned here is truncated and much more can and has been said

- I have had to omit many details
- Other views are available



Reading list on Moodle

- Philosophy: overviews and important texts
- STS studies (mostly physics)
- Non-reading material: inc. CBC podcast – How to Think About Science

Thanks for listening

Post-script

*Some** bits I missed out

*there's much more

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Post-Normal Science

Beyond foundational debates… focus on science in action

Move away from isolated problem-solving experts to groups of interested parties

- Participation of extended peer community: scientists, scholars, industrialists, journalists, campaigners, policy-makers, non-specialist citizens
- Scientific certainty is replaced by extended dialogue across society of risks and benefits
- “Hard facts” replaced by “extended facts” including published results, extended experiences, local surveys and unofficial scientific information
- Organising principle of Quality rather than Truth
- Primacy of Scientific authority replaced by legitimacy of different perspectives from those